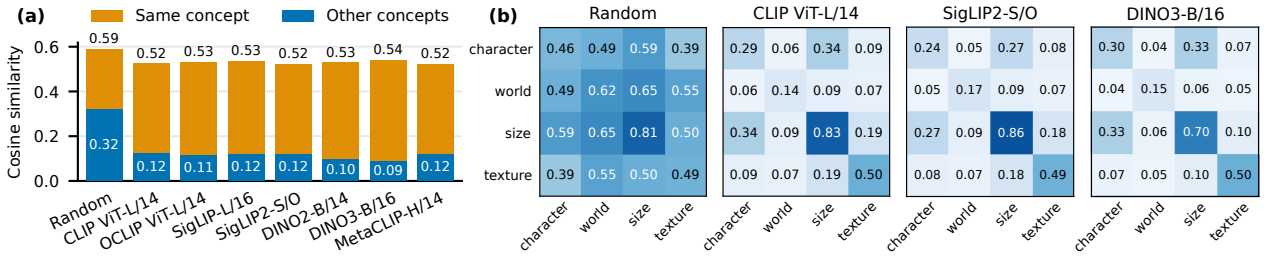

Figure 10. **Pre-trained models exhibit strong within-concept direction similarity and partial orthogonality across concepts. (a)** Aggregated within-concept direction similarity over datasets. **(b)** Pairwise average cosine across concepts. Lower values indicate greater orthogonality between factor vectors. Full orthogonality results are reported in Section I.2.

on the held-out 10% unseen compositions (cf. sampling discussion in Section 4.1). This corresponds to a validity rule $R(T) = 1$ if $|T| = 0.1\,n^k$. We compute *Projected* R^2 on *whitened* $\boldsymbol{P_W} f(\boldsymbol{x})$ (Section 5.1) and pair it with a *compositional accuracy* score on the held-out compositions. We use a randomly-initialized OpenCLIP ViT-L/14 model as a baseline by training linear probes on the embeddings. We use linear probing rather than zero-shot classification to avoid prompt-specification issues, nonetheless, the same conclusions hold in zero-shot experiments on both PUG-Animal and ImageNet-AO (Section I.4).

Compositional accuracy is computed by training one linear classifier per concept, then averaging each classifier's accuracy on the held-out combinations. For example, dSprites has 6 concepts (shape, orientation, x position, y position, size, and color); we train 6 classifiers and report their mean accuracy on unseen combinations.

Results. Across all datasets *higher Projected R^2 coincides with higher compositional accuracy* (Fig. 8). The randomly initialized OpenCLIP ViT-L/14 baseline consistently occupies the low-R^2/low-accuracy corner, indicating the effect is not a dimensionality or scale artifact, and follows the rationale of sanity checks in interpretability (Méloux et al., 2025). The same trend is observed in zero-shot text-probe experiments on both PUG-Animal and ImageNet-AO (Section I.4 and Figs. 28 and 32). A full train/test-split ablation is reported in Section I.1 and Fig. 23. This is consistent with our theory: models whose embeddings are closer to a linear factorization also generalize better to unseen concept combinations.

Takeaway §5.2: Linear factorization in pre-trained models correlates positively with compositional generalization performance.

5.3. Orthogonality of factors

Our theory (Proposition 1) predicts that per-concept difference vectors should be orthogonal *across* concepts under linear factorization, but not necessarily within-concept in generalizing linearly compositional models. We empirically

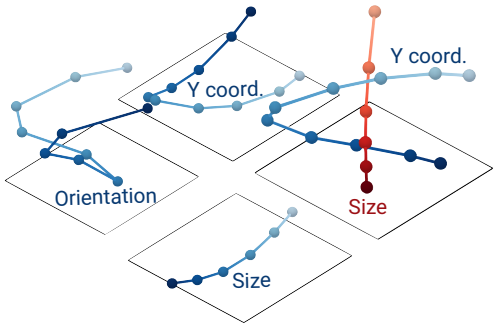

Figure 11. **Geometry of *factors* $\{\boldsymbol{u}_{i,j}\}$ in OpenCLIP ViT-L/14.** The factors are often low dimensional and near co-linear within a concept. Across concepts, the factors are near-orthogonal.

test this prediction by testing orthogonality in two ways: (1) within-concept and (2) across-concept. We defer the details to Appendix I.2.

Results. Pretrained encoders exhibit consistently higher direction similarity within concepts than across concepts (Fig. 10): within-concept similarity (a) is around ≈ 0.53-0.55, whereas cross-concept similarity (b) is ≈ 0.09-0.12. The randomly-initialized encoder also exhibits this pattern; however, the across-concept similarity is higher (0.32 on average) compared to pre-trained models. The same within-vs-across pattern is also observed in zero-shot text-probe experiments on both PUG-Animal and ImageNet-AO (Figs. 29 and 33).

Takeaway §5.3: Pretrained models exhibit partial cross-concept orthogonality, substantially more so than randomly initialized encoders, suggesting that training drives factor directions toward the geometry predicted by our theory.

5.4. Dimensionality of factors

Our theory predicts that generalizing linear compositional models require linear factorization of embeddings into per-concept components. When many concepts must coexist in a fixed embedding dimension, each concept's subspace should be low-rank to enable efficient packing (Section 5.1). Here, we investigate to which extent concept factors in pretrained

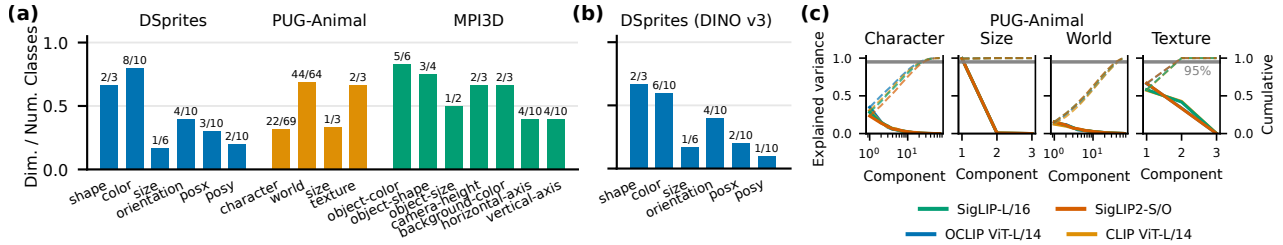

Figure 12. **Dimensionality of factors. (a)** Normalized effective rank across datasets and concepts under OpenCLIP ViT-L/14; text above each bar reports "effective dimension / number of values" for that concept. **(b)** Same analysis as in (a), shown for DINOv3 on dSprites; the recovered factors are typically lower-rank (variance concentrates in fewer PCs). **(c)** Cumulative explained variance of recovered PUG-Animal factors across model families (CLIP, OpenCLIP, SigLIP, SigLIP2); for each concept, the curves are nearly overlapping, indicating strong cross-model similarity in factor geometry.

models are low-dimensional.

Metrics and setup. We study factor geometry (with factor recovery following Section 5.1). For each concept $i \in [k]$ with value set $\mathcal{C}_i$ ($n_i = |\mathcal{C}_i|$), we aggregate the per-concept factors $\boldsymbol{u}_{i,j}$ for $j \in \mathcal{C}_i$ into a matrix $\boldsymbol{U}_i \in \mathbb{R}^{n_i \times d}$. We then analyze (1) the dimensionality of each concept and (2) how this dimensionality compares across models. To do so, we examine the spectrum of $\boldsymbol{U}_i$ (PCA on its rows) and report the number of principal components required to explain 95% of the variance across values j.

Results. Fig. 12 shows that most ordinal factors lie in low-dimensional subspaces relative to their cardinality (e.g., dSprites size $1/6$, MPI3D vertical-axis $4/10$). Panel (b) shows the same tendency in a DINOv3-specific view on dSprites: factor variance typically concentrates in a small number of principal components. Panel (c) shows that this structure is highly consistent across architectures: for each PUG-Animal concept, explained-variance curves from different model families are nearly indistinguishable. This cross-model similarity is consistent with the Platonic Representation Hypothesis (Huh et al., 2024) and with recent evidence that independently trained multimodal contrastive models can often be aligned by near-orthogonal transforms (Gupta et al., 2026). Across datasets and models, $\geq 95\%$ of variance is typically captured by one or two PCs, indicating that spectra align closely by concept. Discrete concepts show higher rank, potentially due to being composed of more atomic attributes. Overall, semantic factors are low-rank and geometrically similar across models, while discrete concepts are not strictly low-rank.

We also visualize dSprites factors (orientation, size, y-position) in Fig. 11. Each subspace is effectively < 3D ($\geq 95\%$ variance in ≤ 2 PCs). Size and y-position trace near-1D path, while orientation forms a smooth 2D curve with small curvature, matching the effective dimensions in Fig. 12.

Takeaway §5.4: Ordinal and continuous factors are typically low-dimensional ($\leq$ 4D), while discrete factors show higher rank, potentially because they encode multiple underlying attributes. Concept factor geometry is similar across different models.

6. Discussion and conclusion

Compositional generalization remains a challenge for vision embedding models, even those trained at scale. In this work we pinned down what compositional generalization demands of a representation. We formalized compositional generalization through three practically motivated desiderata: divisibility (parts must be readable), transferability (readouts trained on a limited but valid subset must generalize), and stability (the learned solution must not depend on which subset of the data space the model was trained on). Under typical training settings, our main theoretical result shows that satisfying these desiderata forces embeddings to behave like a dictionary: representations decompose additively into per-concept factors, and to prevent interference, these factors must be orthogonal across concepts. Importantly, these geometric conditions are also sufficient. However, supporting such a structure is not free: k concepts require at least k dimensions, a bound that in practice may be higher depending on the loss function.

Empirically, across a broad set of modern VLMs and compositional datasets, we find that current representations partially satisfy the predicted structure: additive per-concept components explain a meaningful fraction of variance, cross-concept factors are partially orthogonal, and, notably, the degree to which a model matches the predicted geometry correlates with compositional generalization on unseen combinations. This correlation offers a practical diagnostic for assessing compositional capability directly in representation space. Our factor-rank analysis elucidates how models "pack" many concepts into fixed-dimension embeddings: ordinal and continuous factors tend to be low-dimensional, while discrete factors occupy higher effective rank. These observations hold not only for vision-language models but

also for self-supervised DINO encoders, suggesting the geometric constraints we identify are not an artifact of contrastive language supervision.

Taken together, our results establish that the linear structure widely observed in neural representations is a necessary consequence of compositional generalization. This reframes a large body of empirical findings into a single geometric characterization of what any compositionally generalizing model must achieve. At the same time, current models only partially satisfy these conditions, which may explain their failures on compositional benchmarks. However, as the models continue to scale, these conditions predict the representational geometry they may converge to.

Limitations and future work. Our theory emphasizes worst-case stability over valid training supports; relaxing this to average-case or approximate stability is a natural direction that may better match some practical settings. Characterizing how different training supports change the implied geometry could turn these necessary conditions into a practical design guide for data collection and model building. Our theory assumes a fixed encoder and considers retraining across different training supports; in practice the encoder is trained once on a single dataset; understanding the impact of this assumption on the necessary conditions is an interesting direction for future work.

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